

Doppler Ultrasound Segmentation and Metric Extraction Pipeline

Your Name

July 8, 2026

Contents

1 Introduction

1.1 Motivation and clinical context

1.2 Scope of this document

1.3 Codebase overview

2 Data Inputs and Organisation

2.1 Supported modalities and file formats

2.2 Directory structure and configuration

2.3 Patient grouping and file discovery

3 Preprocessing and Scan Selection

3.1 Scan type detection

3.2 Colour and intensity preprocessing

3.3 Tesseract configuration

4 Text Extraction and OCR Repair

4.1 Metric panel localisation

4.2 Primary OCR pass

4.3 Target label sets

4.4 Label matching algorithms

4.4.1 Exact and subsequence matching

4.4.2 Levenshtein-based fuzzy matching

4.4.3 Ophthalmic-specific matching

4.5 Heart rate refinement

4.6 Metric consistency checks

5 Image Segmentation and Axis Detection

5.1 Initial segmentation of Doppler region

5.2 Axis and tick detection

5.3 Handling invalid or incomplete axes

6 Waveform Digitization

6.1 Ray, morphology, and grow models

6.2 Beat detection and foot-based segmentation

6.3 Time-axis calibration

6.4 Hemodynamic indices

7 Digitized vs OCR Metric Comparison

7.1 Metric mapping into the text table

7.2 Traffic-light scoring

7.3 Model selection